Chapter 2

Sets and Functions

The term 'set' is well known to us. For example, dinner set, set of natural numbers, set of rational numbers etc. As the base of the modern Mathematics set has versatile use. The German mathematician George Cantor (1845-1918) first explained the concept of set. He introduced the notion of infinite set, and created a sensation in Mathematical science. His notion is now known as set theory. In this chapter, we shall solve different problems by using concepts of set, diagrams and logic. Again idea of function will also be introduced.

After completion of this chapter, the learners will be able to–

- ► express the concepts of sets and subsets through symbols;
- ► describe how sets can be expressed;
- ► define infinite sets, and distinguish between finite and infinite sets;
- ► define and verify union and intersection of sets;
- ► define power sets, and construct power sets of sets with two or three elements;
- ► define ordered pairs and cartesian products;
- ► prove simple set theoretic rules by using examples and Venn diagrams, and apply these rules for solving different problems;
- ► define relations and functions, and construct them.
- ► determine domain and range;
- ► draw diagrams for functions.

Sets

A set is simply a well organized object of real or imaginary world. For example, the set of three text books of Bangla, English and Mathematics. Some other

examples are the set of first 10 odd natural numbers, set of integers, set of real numbers, etc. Sets are usually denoted by the capital letters of English alphabet like $A, B, C, \ldots X, Y, Z$.

For example, the set of three numbers $2, 4, 6$ can be denoted by $A = \{2, 4, 6\}$

Each object or member of the set is called its element. For example, if $B = \{a, b\}$, to express that a and b are its elements the sign $\in$ is used to express the element. a and b are elements of B;
$\therefore a \in B$ and we can read as a, is a member of B (a belongs to B)

$b \in B$ is a member of b, (b belongs to B)

The above set B does not contain c. So it is expressed as $c \notin B$ and is read as c is not a member of B (c does not belong to B)

Methods for expressing sets

Sets can be expressed using two methods, namely Roster method or tabular method and Set Builder Method.

Roster method: In this method all members are listed and put inside {} and if there are more than one element, then elements are separated by commas. For example, $A = \{a, b\}$, $B = \{2, 4, 6\}$, $C = \{$Niloy, Shuvro, Tisha$\}$, etc.

Set builder method: In this method all elements are not listed rather their common property is described. For example, $A = \{x : x$ is an odd natural number$\}$, $B = \{x : x$ is the first five students of class IX $\}$ etc. Here, by ':' has been expressed as 'such that'. Since in this method, element of the set is specified by rules, this method is called **Rule Method**.

Example 1. Express $A = \{7, 14, 21, 28\}$ in set building method.

Solution: The elements of A are $7, 14, 21, 28$.

Here, each element is a multiple of 7 and not exceeding 28.

$\therefore A = \{x : x|,$ is a multiple of 7 and $0 < x \leq 28\}$

Example 2. Express $B = \{x : x,$ is a divisor of $28\}$ by using roster method.

Solution: Here, $28 = 1 \times 28 = 2 \times 14 = 4 \times 7$

$\therefore$ factors of 28 are $1, 2, 4, 7, 14, 28$

Hence the desired set is $B = \{1, 2, 4, 7, 14, 28\}$

Example 3. Express $C = \{x : x \text{ positive integer and } x^2 < 18\}$ by using roster method.

Solution: $1, 2, 3, 4, 5, \ldots$ are positive integers.

Here, if

$x = 1$, then $x^2 = 1^2 = 1$; if $x = 2$, then $x^2 = 2^2 = 4$

If $x = 3$, then $x^2 = 3^2 = 9$; if $x = 4$, then $x^2 = 4^2 = 16$

If $x = 5$, then $x^2 = 5^2 = 25$; which is greater than 18.

$\therefore$ as per conditions the acceptable positive integers are $1, 2, 3$ and 4.

$\therefore$ the given set is $C = \{1, 2, 3, 4\}$

Work:

1) Express $C = \{-9, -6, -3, 3, 6, 9\}$ by using set building method.
2) Express $B = \{y : y \text{ integer and } y^3 \leq 18\}$ by using set building method.

Finite Set

The set number of elements of which can be determined by counting is called finite set. For example, $D = \{x, y, z\}$, $E = \{3, 6, 9, \ldots, 60\}$, $F = \{x : x \text{ prime and } 30 < x < 70\}$, etc. are finite sets. Here, set D contains 3 elements, set E 20 elements and set F 9 elements.

Infinite Set

The set elements of which cannot be expressed by counting is called infinite set. For example, $A = \{x : x \text{ odd natural number}\}$, set of natural numbers $N = \{1, 2, 3, 4, \ldots\}$, set of integers $Z = \{\ldots, -3, -2, -1, 0, 1, 2, 3 \ldots\}$, set of rational numbers $Q = \left\{\frac{a}{b} : a \text{ and } b \text{ integer and } b \neq 0\right\}$, set of rational numbers R etc. are infinite sets.

Example 4. Prove that the set of all natural numbers is an infinite set.

Solution: Suppose the set of natural numbers N is a finite set. Then this set must have a maximum element K, where K∈N. However, according to the concept of natural numbers, if K is a natural number, then K+1 is also a natural number which is greater than K. Therefore, K+1 must be an element of the set of natural numbers N, i.e., K+1∈N.

But initially, we assumed K as the maximum element of the set N. Subsequently, it turns out that K+1 is also an element of the set N. Similarly, it can be shown that K+2, K+3, ... are also elements of the set N.
Therefore, the set of natural numbers N cannot be finite. Hence, the set of natural numbers is an infinite set.

Work: Determine which ones are finite and which ones are infinite sets:

1) $\{3, 5, 7\}$
2) $\{1, 2, 2^2, \ldots 2^{10}\}$
3) $\{3, 3^2, 3^3, \ldots\}$
4) $\{x : x \text{ integer and } x < 4\}$
5) $\{\frac{p}{q} : p \text{ and } q \text{ are mutually coprimes and } q > 1\}$
6) $\{y : y \in N \text{ and } y^2 < 100 < y^3\}$

Empty Set

If a set does not have any element, the set is said to be an empty set, and is denoted by $\varnothing$. For example, the set of three male students of Girl's school, $\{x \in N : 10 < x < 11\}$, $\{x \in N : x \text{ prime and } 23 < x < 29\}$ etc.

Venn-Diagram

John Venn (1834-1923) first expressed sets by using pictures. Here every set is represented by a geometric figure like rectangles, cicrles and triangles in a plane. These pictures have been named after John Venn.

Subset

$A = \{a, b\}$ is a set. Sets $\{a, b\}$, $\{a\}$, $\{b\}$ can be formed from the elements of this set. Again, we can form a subset $\varnothing$ with no elements at all. These sets $\{a, b\}$, $\{a\}$, $\{b\}$, $\varnothing$ are subsets of set A. All the sets that can be formed from a set are called subsets of the original set. Subset is denoted by $\subseteq$. If B is a subset of A, then we write $B \subseteq A$. B is a subset of A. Among the subsets above $\{a, b\}$ is equal to set A. Each set is the subset of itself. again $\varnothing$ in formed from any set. $\therefore \varnothing$ is a subset of any set.

Let $P = \{1, 2, 3\}$ and $Q = \{2, 3\}$, $R = \{1, 3\}$,then P, Q and R are subsets of P. That is, $P \subseteq P$, $Q \subseteq P$ and $R \subseteq P$.

Proper Subset

Of the subsets constructed from a set, the ones that have lesser number of elements than the original set is called proper set. For example, $A = \{3, 4, 5, 6\}$ and $B = \{3, 5\}$ are two sets. Here all elements of B are also elements of A, and number of elements in B is less than that of A.

$\therefore$ B is a proper subset of A and is expressed as $B \subset A$.

In the examples of subsets Q and R are each proper subset of P. It may be mentioned here that $\varnothing$ is a proper subset of any set.

Example 5. Write down the subsets of $P = \{x, y, z\}$ and identify those that are proper subsets.

Solution: Given, $P = \{x, y, z\}$

Subsets of P are $\{x, y, z\}, \{x, y\}, \{x, z\}, \{y, z\}, \{x\}, \{y\}, \{z\}, \varnothing$.

Proper subsets of P are $\{x, y\}, \{x, z\}, \{y, z\}, \{x\}, \{y\}, \{z\}, \varnothing$

Observation: If number of elements of a set is n, then it has 2^n subsets, and number of proper subsets is $2^n - 1$.

Equivalent Set

If two sets have the same elements, then they are said to be equivalent sets. For example, $A = \{3, 5, 7\}$ and $B = \{5, 3, 3, 7\}$ are two equivalent sets, then they are denoted by $A = B$. Note that $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$

Again, if $A = \{3, 5, 7\}$, $B = \{5, 3, 3, 7\}$ and $C = \{7, 7, 3, 5, 5\}$, then sets A, B and C are equivalent sets. That is $A = B = C$

Observation: If order of elements in the set are changed or certain element is repeated set is not changed.

Difference of Sets

The set obtained by removing the elements of the second set from the elements of the first set is called the difference of the two sets. The difference of two sets, A and B, is written as $A \setminus B$ or $A - B$ (read as 'A minus B'). It is defined as:

$A - B = \{x : x \in A, x \notin B\}$

This means that the resulting set consists of all elements (x) that are in set A but are not in set B.

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 5\}$. If we delete the elements of set B from that of A, then we get $\{1, 2, 4\}$.

$\therefore$ $A - B = \{1, 2, 3, 4, 5\} - \{3, 5\} = \{1, 2, 4\}$

Example 6. If $P = \{x : x$ is a factor of 12 $\}$ and $Q = \{x : x$ is a multiple of 3 and $x \leq 12\}$, then determine $P - Q$.

Solution: Given , $P = \{x : x \text{ is a factor of } 12\ \}$

Here, factors of 12 are $1, 2, 3, 4, 6, 12$

$\therefore P = \{1, 2, 3, 4, 6, 12\}$

Again, $Q = \{x : 3 \text{ is a factor of } x \text{ and } x \leq 12\}$

Here, multiples of 3 up to 12 are $3, 6, 9, 12$

$\therefore Q = \{3, 6, 9, 12\}$

$\therefore P - Q = \{1, 2, 3, 4, 6, 12\} - \{3, 6, 9, 12\} = \{1, 2, 4\}$

The required set is: $\{1, 2, 4\}$

Universal Set

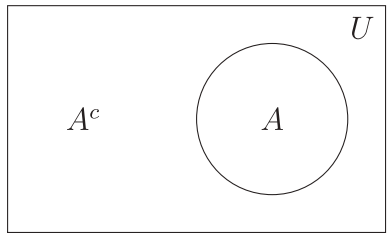
All sets under discussion are subsets of a definite set. For example: the set $A = \{x, y\}$ is a subset of the set $B = \{x, y, z\}$ Here, B set is called as universal set with respect to set A.

So if all sets under discussion are subsets of a given set, then the later set is said to be **universal set** with respect to sets under discussion.

Usually universal set is denoted by U. However, other symbols can also be used to denote universal set. For example, if set of all even natural numbers is $C = \{2, 4, 6, \ldots\}$ and set of all natural numbers is $N = \{1, 2, 3, 4, 5, 6, \ldots\}$, then N will be universal set with respect to the set C.

Complement of a Set

Let U be universal set and A be a subset of U. Then the set of elements outside of A is termed as **complementary set** of A. The complementary set of A is denoted by A^c or A'. Mathematically, $A^c = U \setminus A$.



Let, P and Q be two sets. Then the elements of P that are not elements of Q are said to form the complementary set of P with respect to Q, and is written $Q^c = P \setminus Q$

Example 7. If $U = \{1, 2, 3, 4, 5, 6, 7\}$, $A = \{2, 4, 6, 7\}$ and $B = \{1, 3, 5\}$, then determine A^c and B^c.

Solution: $A^c = U \setminus A = \{1, 2, 3, 4, 5, 6, 7\} \setminus \{2, 4, 6, 7\} = \{1, 3, 5\}$

and $B^c = U \setminus B = \{1, 2, 3, 4, 5, 6, 7\} \setminus \{1, 3, 5\} = \{2, 4, 6, 7\}$

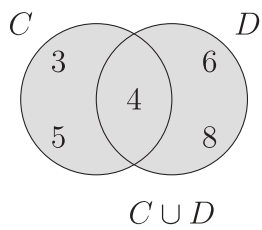
The required sets are $A^c = \{1, 3, 5\}$ and $B^c = \{2, 4, 6, 7\}$

Union of Sets

The set consisting of elements of two or more sets is said to be **union of sets.** Let, A and B be two sets. The union of sets A and B is denoted by $A \cup B$ and read as A union B. In set building method $A \cup B = \{x : x \in A \text{ or } x \in B\}$

Example 8. If $C = \{3, 4, 5\}$ and $D = \{4, 6, 8\}$, determine $C \cup D$.

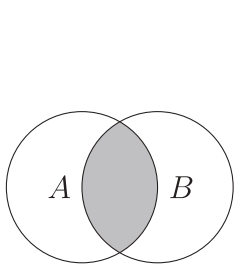
Solution: Given $C = \{3, 4, 5\}$
and $D = \{4, 6, 8\}$
$\therefore C \cup D = \{3, 4, 5\} \cup \{4, 6, 8\} = \{3, 4, 5, 6, 8\}$
The set to be determined is: $\{3, 4, 5, 6, 8\}$



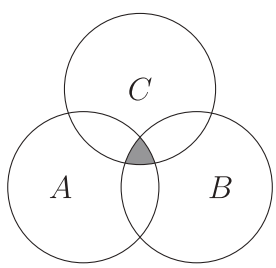
$C \cup D$

Intersection of Sets.

The set consisting of common elements of two or more sets is known as **intersection set** Let A and B be two sets. The intersection of sets A and B is denoted by $A \cap B$ and read as A intersection B. In set building notation $A \cap B = \{x : x \in A \text{ and } x \in B\}$



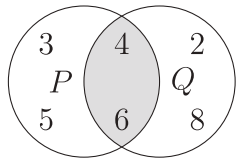
$A \cap B$



$A \cap B \cap C$

Example 9. If $P = \{x \in N : 2 < x \leq 6\}$ and $Q = \{x \in N : x \text{ is even and } x \leq 8\}$, determine $P \cap Q$.

Solution: Given,
$P = \{x \in N : 2 < x \leq 6\} = \{3, 4, 5, 6\}$
$Q = \{x \in N : x \text{ is even and } x \leq 8\} = \{2, 4, 6, 8\}$
$\therefore P \cap Q = \{3, 4, 5, 6\} \cap \{2, 4, 6, 8\} = \{4, 6\}$
The set to be determined is $\{4, 6\}$



$P \cap Q$

Disjoint Set

If there is no element common to any two sets, then these two sets are called **disjoint sets.** Let A and B be two sets. If $A \cap B = \varnothing$ then A and B are said to be mutually disjoint sets. Example: there is no common element of the sets $A = \{1, 2, 3\}$ and $B = \{4, 5\}$. Thus A and B are two mutually disjoint sets.

Work: If $U = \{1, 3, 5, 9, 7, 11\}$, $E = \{1, 5, 9\}$ and $F = \{3, 7, 11\}$, then determine $E^c \cup F^c$ and $E^c \cap F^c$.

Power Sets

$A = \{m, n\}$ is a set. Subsets of A are $\{m, n\}, \{m\}, \{n\}, \varnothing$, and set of all sets $\{\{m, n\}, \{m\}, \{n\}, \varnothing\}$ of A is called **power set** of A. Power set of A is denoted by $P(A)$. So the set formed by all subsets of a set is called power set of a given set.

Example 10. Find the number of elements of the power sets $A = \varnothing$, $B = \{a\}$, $C = \{a, b\}$.

Solution: Here, $P(A) = \{\varnothing\}$.

$\therefore$ number of elements of A is 0 and number of elements of its power set is $= 1 = 2^0$

Again $P(B) = \{\{a\}, \varnothing\}$

$\therefore$ number of elements of B is 1, and number of eleemnts of its power set is $= 2 = 2^1$.

and $P(C) = \{\{a\}, \{b\}, \{a, b\}, \varnothing\}$

$\therefore$ number of elements of C is 2 and number of elements of its power set is $= 4 = 2^2$

So, if number of elements of a set is n, then number of elements of its power set is 2^n.

Work: If $G = \{1, 2, 3\}$,then determine $P(G)$. Show that number of elements of $P(G)$ is 2^3

Ordered Pair

Amena and Sumena of class VIII occupied 1st and 2nd position repectively in the annual examination. According to merit this can be expressed as (Amena, Sumena) pair. Similar definite pairs are called ordered pairs.

So, for a pair of elements if the positions are given and expressed, then the pair is called **Ordered Pair.**

If the first element of an ordered pair is x, and the second element is y, then the ordered pair is denoted by (x, y). For the ordered pairs (x, y) and (a, b) will the equal or $(x, y) = (a, b)$ if and only if $x = a$ and $y = b$.

Example 11. If $(2x + y, 3) = (6, x - y)$ then,determine (x, y).

Solution: Given,$(2x + y, 3) = (6, x - y)$

As per conditions of ordered pairs,

$2x + y = 6 \cdots\cdots (1)$

$x - y = 3 \cdots\cdots (2)$

By adding equations (1) and (2) we get $3x = 9$ or $x = 3$

Putting value of x in equation (1) we get, $6 + y = 6$ or $y = 0$

$\therefore (x, y) = (3, 0)$

(Cartesian Product)

Mr Karim decided to colour interior wall of his rooms using white or blue colour, and outside wall using red, yellow or green. The set of colours of interior wall is $A = \{$white, blue$\}$ and that of outside wall is $B = \{$red, yellow and green $\}$ Mr Karim can colour walls of his rooms by (white, red), (while, yellow), (white, green), (blue, red), (blue, yellow), (blue, green) ordered pairs. The above ordered pairs are writtena s below:

$A \times B = \{$(white, red), (while, yellow), (white, green), (blue, red), (blue, yellow), (blue, green)$\}$

The above set of ordered pairs is called **Cartesian Product set.**

In set building method, $A \times B = \{(x, y) : x \in A \text{ and } y \in B\}$

$A \times B$ is read as A cross B.

Example 12. If $P = \{1, 2, 3\}$, $Q = \{3, 4\}$, $R = P \cap Q$,then determine $P \times R$ and $R \times Q$.

Solution: Given, $P = \{1, 2, 3\}$, $Q = \{3, 4\}$

and $R = P \cap Q = \{1, 2, 3\} \cap \{3, 4\} = \{3\}$

$\therefore P \times R = \{1, 2, 3\} \times \{3\} = \{(1, 3), (2, 3), (3, 3)\}$

and $R \times Q = \{3\} \times \{3, 4\} = \{(3, 3), (3, 4)\}$

Work:

1) If $\left(\frac{x}{2} + \frac{y}{3}, 1\right) = \left(1, \frac{x}{3} + \frac{y}{2}\right)$,then determine (x, y).
2) If $P = \{1, 2, 3\}$, $Q = \{3, 4\}$ and $R = \{x, y\}$,then determine $(P \cup Q) \times R$ and $(P \cap Q) \times Q$.

Example 13. Determine the set of natural numbers that have the same residue 23 when they divide 311 and 419.

Solution: The natural numbers that leave 23 as residue when they divide 311 and 419 will be bigger than 23 and will be common factors of both $311 - 23 = 288$ and $419 - 23 = 396$.

Let the set of factors of 288 are larger than 23 be A.

Here $288 = 1 \times 288 = 2 \times 144 = 3 \times 96 = 4 \times 72 = 6 \times 48 = 8 \times 36 = 9 \times 32 = 12 \times 24 = 16 \times 18$

$\therefore A = \{24, 32, 36, 48, 72, 96, 144, 288\}$

Let the set of factors of 396 greater than 23 be B.

Here $396 = 1 \times 396 = 2 \times 198 = 3 \times 132 = 4 \times 99 = 6 \times 66 = 9 \times 44 = 11 \times 36 = 12 \times 33 = 18 \times 22$

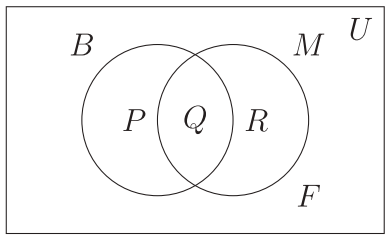
$\therefore B = \{33, 36, 44, 66, 99, 132, 198, 396\}$

$\therefore A \cap B = \{24, 32, 36, 48, 72, 96, 144, 288\} \cap \{33, 36, 44, 66, 99, 132, 198, 396\}$

$\therefore A \cap B = \{36\}$

The required set is $\{36\}$

Example 14. Of 100 students 88 passed in Bangla, 80 in Mathematics and 70 in both subjects. Express the information in Venn diagram, and find how many students failed in both subjects.



Solution: In the Venn diagram rectangular area represents U the set of 100 students. The set of students who passed in Bangla and Mathematics are denoted respectively by B and M. As a result the Venn diagram is divided into 4 disjoint sets denoted by P, Q, R and F.

Here, the set of students passed in both subjects is $Q = B \cap M$, cardinality of which is 70.

P is the set of students passed in Bangla only. Its cardinality is $|P| = 88 - 70 = 18$.

R is the set of students who passed only in Mathmatics. $\therefore \quad |R| = 80 - 70 = 10$

Students who passed in at least one subject is $P \cup Q \cup R = B \cup M$, and corresponding cardinality is $18 + 10 + 70 = 98$

F is the set of students who failed in both the subjects, and $|F| = 100 - 98 = 2$

$\therefore$ number of students failed in both the subjects is 2.

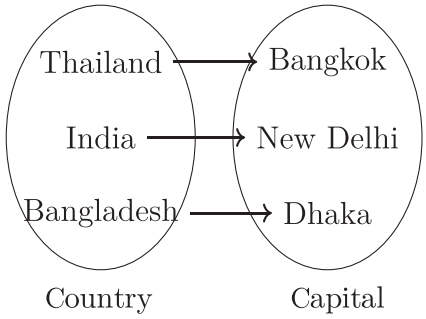
Exercises 2.1

1. Express the following sets in the set builder notation:
   1) $\{x \in N : x^2 > 9 \text{ and } x^3 < 130\}$
   2) $\{x \in Z : x^2 > 5 \text{ and } x^3 \leq 36\}$
   3) $\{x \in N : x \text{ is factor of } 36 \text{ and } x \text{ is a multiple of } 6\}$
   4) $\{x \in N : x^3 > 25 \text{ and } x^4 < 264\}$
2. Express the following sets in the set builder notation:
   1) $\{3, 5, 7, 9, 11\}$
   2) $\{1, 2, 3, 4, 6, 9, 12, 18, 36\}$
   3) $\{4, 8, 12, 16, 20, 24, 28, 32, 36, 40\}$
   4) $\{\pm 4, \pm 5, \pm 6\}$
3. If $A = \{2, 3, 4\}$, $B = \{1, 2, a\}$ and $C = \{2, a, b\}$,then determine the following sets:
   1) $B \setminus C$ 2) $A \cup B$ 3) $A \cap C$
   4) $A \cup (B \cap C)$ 5) $A \cap (B \cup C)$
4. If $U = \{1, 2, 3, 4, 5, 6, 7\}$, $A = \{1, 3, 5\}$, $B = \{2, 4, 6\}$ and $C = \{3, 4, 5, 6, 7\}$, verify the following statements:
   1) $(A \cup B)' = A' \cap B'$
   2) $(B \cap C)' = B' \cup C'$
   3) $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$
   4) $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$
5. If $Q = \{x, y\}$ and $R = \{m, n, l\}$,then determine $P(Q)$ and $P(R)$.
6. If $A = \{a, b\}$, $B = \{a, b, c\}$ and $C = A \cup B$, then show that number of elements of $P(C)$ is 2^n, where n is the number of elements of C.
7. 1) If $(x - 1, y + 2) = (y - 2, 2x + 1)$,then determine x and y.
   2) If $(ax - cy, a^2 - c^2) = (0, ay - cx)$,then determine (x, y).
   3) If $(6x - y, 13) = (1, 3x + 2y)$,then determine (x, y).
8. 1) If $P = \{a\}$, $Q = \{b, c\}$,then determine $P \times Q$ and $Q \times P$.
   2) If $A = \{3, 4, 5\}$, $B = \{4, 5, 6\}$ and $C = \{x, y\}$, determine $(A \cap B) \times C$.
   3) If $P = \{3, 5, 7\}$, $Q = \{5\ 7\}$ and $R = P \setminus Q$, determine $(P \cup Q) \times R$.

9. If A and B are respectively sets of all factors of 35 and 45, then determine $A \cup B$ and $A \cap B$.
10. Determine the natural numbers that divide both 346 and 556 to have 31 as residue.
11. Of the 30 students in a class, 20 like football and 15 cricket. Number of students who like both the games is 10. Using Venn diagram determine the number of students who do not like either of the games.
12. Of 100 students 65 passed in Bangla, 48 in both Bangla and English and 15 failed in both the subjects.
    1) If $(1,\ x+y) = (2x-y,\ 5)$, then find $(x,\ y)$.
    2) Determine the number of students who passed only in Bangla or passed only in English.
    3) Determine union of the sets of prime factors of number of students passed and failed in both subjects.

Relations

We know Dhaka is the capital of Bangladesh, New Delhi is of India and Bangkok is of Thailand. There is a relation between capitals and countries. This is country-capital relation. This relation can be expressed as a set in the following way:



That is country-capital relation = {(Bangladesh, Dhaka), (India, New Delhi), (Thailand, Bangkok)}.

If A and B are two sets then, nonnull subset R of ordered pairs of their Cartesian product $A \times B$ is said to be a relation from the set A to the set B. Here set R is a subset of the set $A \times B$. That is, $R \subseteq A \times B$